

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Tech. (CE/IT/EC) Examination

May 2014

CE104 Object Oriented Programming with C++**Date: 19.05.2014, Monday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I**Q - 1 Do as directed.****(a) Fill in the Blank.****[03]**

1. Instance of a class is called _____.
2. Class members are _____ by default.
3. A class which cannot be used to create object is called _____.

(b) State True or False.**[02]**

1. It is not valid to use variables of different data types in same arithmetic expression.
2. 'inline' keyword is request only to compiler, it may not be granted.

(c) Why are the words cout and cin not considered as keywords?**[01]****(d) Define Encapsulation.****[01]****(e) Find the output of following code.****[04]**

```
1. #include<iostream>
   using namespace std;
```

```
int main()
{
    int a, b = 10;
    a = -b--;
    cout<<"a="<<a<<"b="<<b;
    return 0;
}
```

```
2. #include<iostream>
   using namespace std;
   class A{
       public:
       virtual void fun()
       {cout << "A" << endl;}
```

```
};
   class B: public A{
       public:
       virtual void fun()
       {cout << "B" << endl;}
```

```
};
   class C: public B{
       public:
       virtual void fun()
       {cout << "C" << endl;}
```

```
};
int main(){
    B b;
    C c;
    A *p1=&b, *p2=&c;
    p1->fun();
    p2->fun();
    return 0;
}
```

Q - 2 Answer the following questions.

- (a) What is *operator* overloading? Explain with example overloading of unary operator? [05]
- (b) A class with two data members and two member functions is given. Explain memory allocation of objects of the class with diagram. [05]
- (c) List out the characteristics of *friend* function. [02]

OR

Q - 2 Answer the following questions.

- (a) What is destructor? Explain in detail with appropriate example. [05]
- (b) List out the basic concepts of object oriented programming. Explain any three in detail. [05]
- (c) Explain following function with respect to files. [02]
 - 1. seek()
 - 2. eof()

- Q-3 (a)** Write a program to create a class **Medicine** which stores type of medicine, name of company, date of manufacturing. Class **Tablet** is inherited from **Medicine**. **Tablet** class has name of tablet, quantity per pack, price of one tablet as members. Class **Syrup** is also inherited from **Medicine** and it has quantity per bottle, dosage unit as members. Both the classes contain necessary member functions for input and output data. Write a `main( )` that enter data for tablet and syrup, also display the data. [06]
- (b) Define a class **NUM** having two integer members `a` and `b`. A class has member functions for input and output. Overload the operators `-` and `==` such that it supports the following statements: [06]
- ```
NUM N1, N2, N3; N1=N2-N3; NUM N4,N5; if (N4==N5) {...}
```

**OR**

- Q-3 (a)** Write a program to open a file, count number of alphabets in the file and display the count. [06]
- (b) Define a class **Distance** having integer members `cm` and `mm`. A class has member functions to enter and display data. Also write appropriate member functions such that it supports the following statements. [06]
- ```
Distance D1, D2; D2 = 24; Here D2 gets 2 cm and 4 mm.
```

SECTION - II

Q - 4 Attempt the following questions.

- (a)
 - 1. List out the *operators* which cannot be overloaded in C++.
 - 2. Differentiate between procedure oriented programming and object oriented programming.
 - 3. Explain *reference* variable.[06]
- (b) **Fill in the blanks with most appropriate words.** [03]
 - 1. _____ is used to convert class type data to basic type.
 - 2. _____ pointer is automatically passed to a member function when it is called.
 - 3. A _____ is a sequence of bytes.

(c) Give the output of following code.

[02]

```
#include<iostream>
using namespace std;
void manip(int &b,int c=100)
{
    int temp=b+c;
    b+=temp;
    if(c==100)
        cout<<temp<<" "<<b<<endl;
}
int main()
{
    int m=90, n=10;
    manip(m);
    cout<<m<<" "<<n<<endl;
    return 0;
}
```

Q - 5 Answer the following questions.

- (a) In which situation virtual base class is required. Explain with example. [05]
- (b) Explain the concept of pointer to object with example. [05]
- (c) Draw and explain general structure of a typical C++ program. [02]

OR

Q - 5 Answer the following questions.

- (a) Define constructor. List out characteristics of constructor. Differentiate between default constructor and parameterized constructor. [05]
- (b) What is inheritance? List out different forms of inheritance. Explain multilevel inheritance with example. [05]
- (c) Differentiate between function overloading and function overriding. [02]

Q -6 (a) Write a program using the concept of *function overloading* to calculate volume of cube= $(side \times side \times side)$, cylinder= $(3.1428 \times r^2 \times h)$ and rectangular = $(l \times b \times h)$. Input the relevant values and calculate the volume of each shape. [06]

- (b) Create a class **Amount** having members: rupees and paisa in integer. Define a default constructor. Also define parameterized constructor to accept amount in Paisa and initialize it to rupees and paisa. Define another member function to display amount in Rs::ps format. Write a main() to demonstrate the class. [06]
For example, 120 Paisa = 1 rupee and 20 paisa.

OR

Q -6 (a) Write a program for 3×3 matrix in which program should read two 3×3 matrices, perform addition of two and display the final matrix. [06]

- (b) Declare a class **TIME** having members like hour, minutes and seconds. TIME class has functions to input and display value. Time will be displayed in HH:MM:SS format. TIME has one more function update which will increment the time by one second. Write a main function to create time object, initialize it, update it and display updated time. [06]
